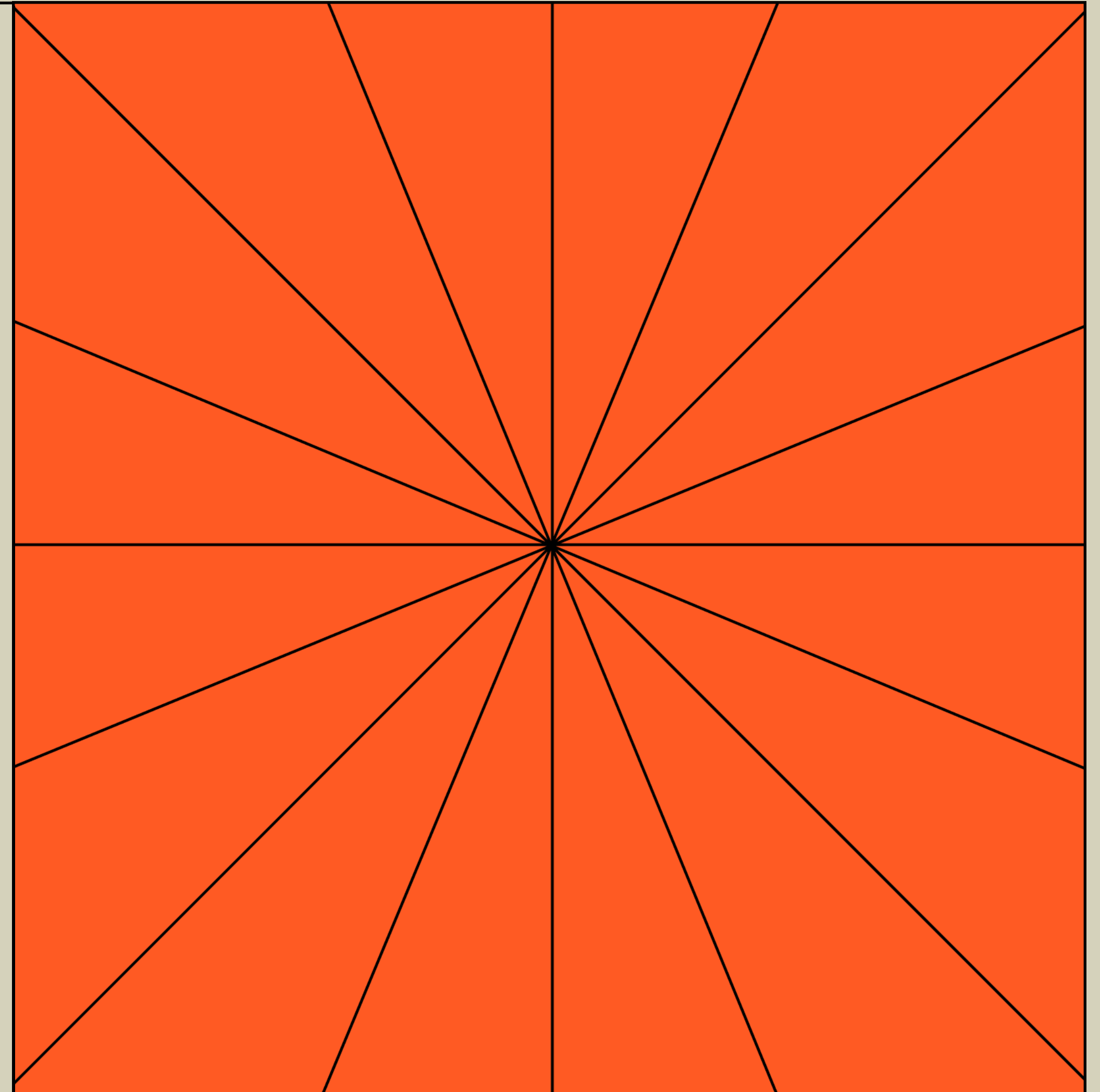
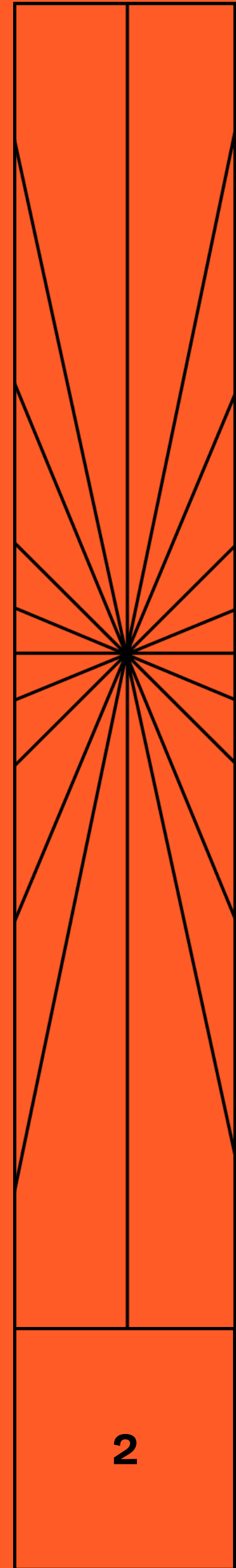


NOISE-AWARE ANSATZ DESIGN

For Digitized Adiabatic Quantum Computing
on IBM Heron Hardware



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TL;DR

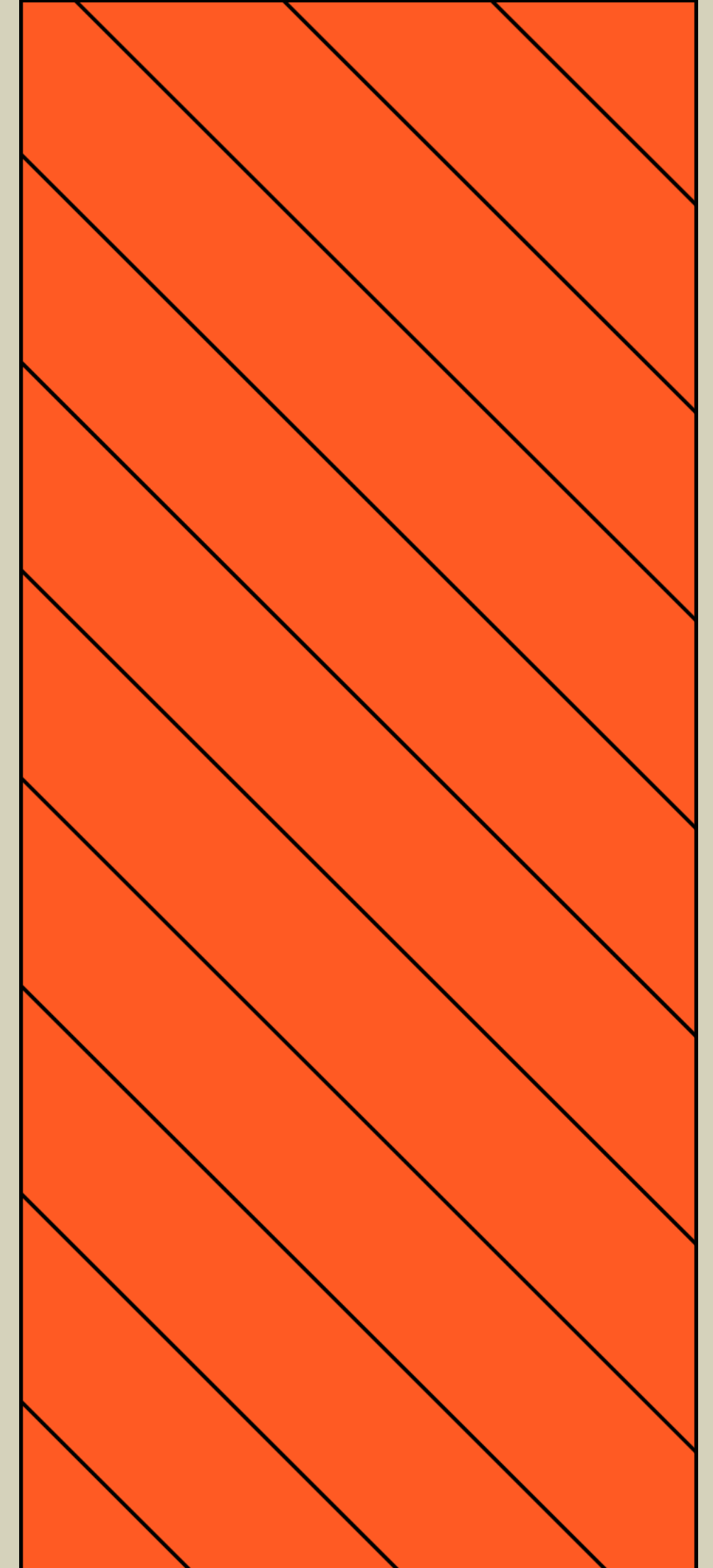
We design and test a noise-aware ansatz for finding the ground state of the Transverse-Field Ising Model (TFIM) on IBM Heron hardware.

We vary three circuit design dimensions: the **interpolation schedule**, the **trotter structure**, and **counterdiabatic driving**.

THE TRANSVERSE-FIELD ISING MODEL (TFIM)

$$H(t) = (1 - t) H_0 + t H_1 \quad H_0 = - \sum_i X_i \quad H_1 = - \sum_i Z_i Z_{i+1}$$

- Starting State ($t = 0$): Pure transverse field H_0 . Ground state is uniform superposition
- Target State ($t = 1$): Pure Ising coupling H_1
 - Degenerate ground space, adiabatic evolution selects GHZ state
 - Ground energy = $-J(N-1)$
- Critical Point, Quantum phase transition occurs at $t_c = \frac{1}{1 + J}$



HERON R2

149.77

T2 (median micro seconds)

266.59

T1 (median micro seconds)

ID, SX, X, CZ

Fixed pulses

RZZ(θ) RX(θ)

Parameterised pulses

RZ(θ)

Virtual gate

1.83E-3

2Q error (median)

9.22E-3

Readout error (median)

THE TROTTER ERROR VS GATE NOISE TRADE-OFF

TROTTER ERROR

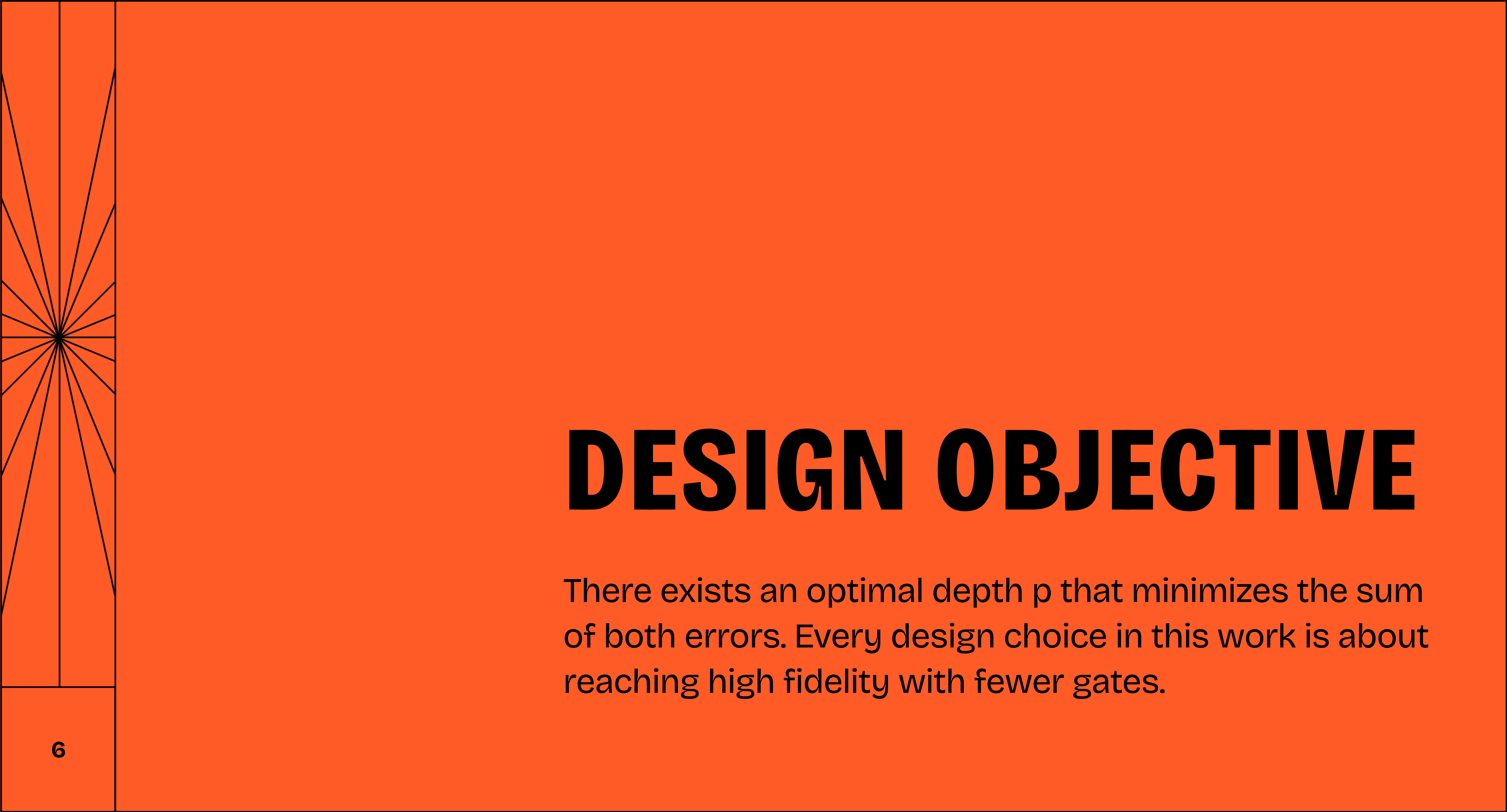
Sequential application of non-commuting operators is an approximation. Error shrinks as $1/p^2$ (Suzuki-Trotter) and $1/p$ (Lie-Trotter)
— more steps means better accuracy.

TOO SHALLOW

GATE NOISE

Every RZZ gate has probability $2Q_error$ of introducing an error. Errors accumulate as $(1 - 2Q_error)^{n_gates}$
— more steps means lower fidelity.

TOO DEEP



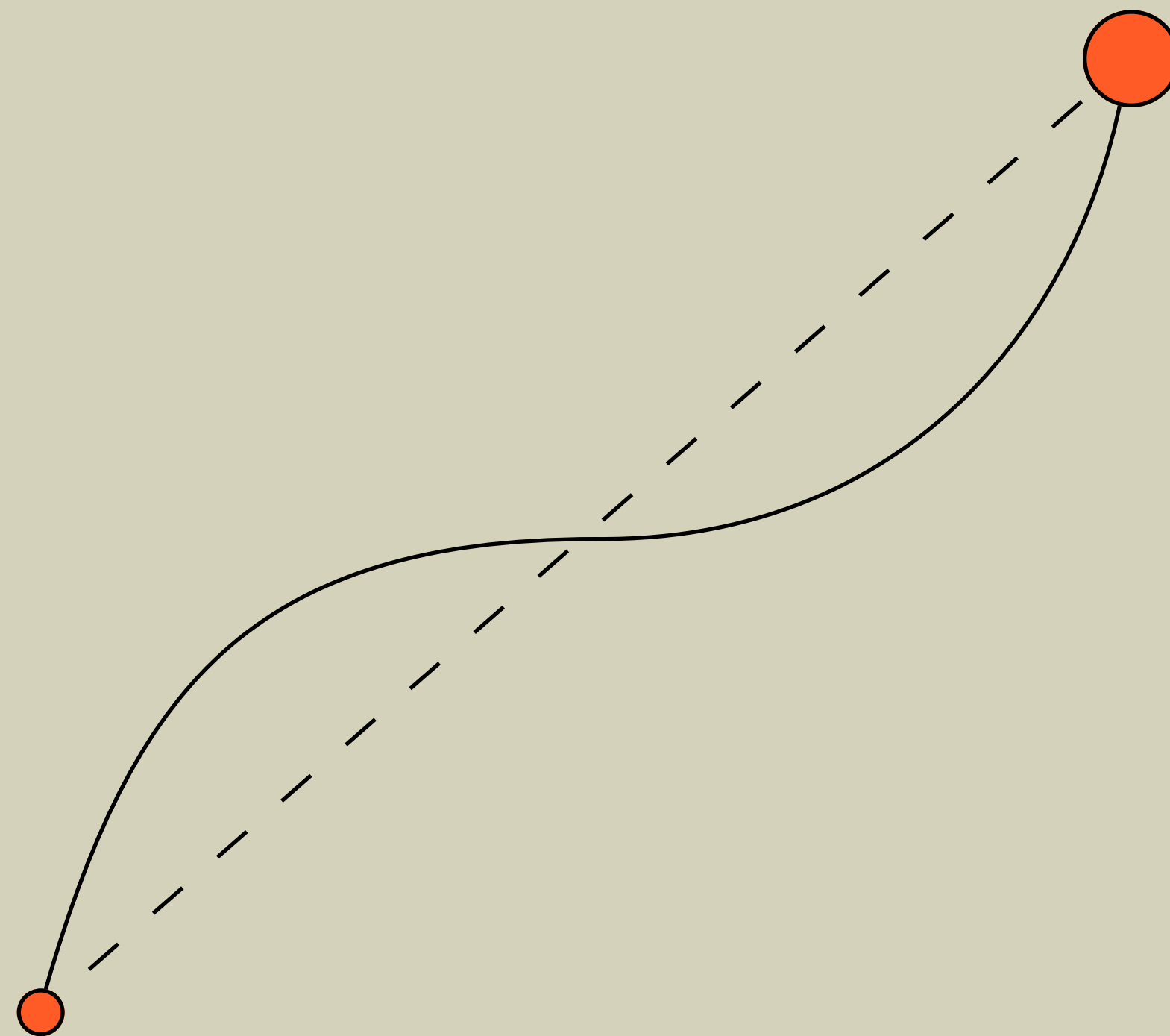
DESIGN OBJECTIVE

There exists an optimal depth p that minimizes the sum of both errors. Every design choice in this work is about reaching high fidelity with fewer gates.

THE INTERPOLATION SCHEDULE

Why not just sweep at a constant rate?

A linear schedule treats every part of the adiabatic path equally — but not every part is equally hard.



- Required evolution time $\sim 1 / \text{gap}^2$ [1]
- Gap is smallest at the quantum critical point
- We use a critical-point-aware schedule that slows the sweep through the quantum critical point
- For $t_c = 0.5$, this is implemented as the trigonometric form:
$$s(t) = t + \frac{c}{2\pi} \sin(2\pi t)$$
- Minimum gap shrinks as $1/N$ — harder with more qubits [2]

[1] Farhi, E., Goldstone, J., Gutmann, S., & Sipser, M. (2000). Quantum computation by adiabatic evolution. arXiv:quant-ph/0001106. MIT-CTP-2936.

[2] Farhi, E., Goldstone, J., Gosset, D., Gutmann, S., Meyer, H. B., & Shor, P. (2010). Quantum adiabatic algorithms, small gaps, and different paths. arXiv:0909.4766v2 [quant-ph]. MIT-CTP 4076 / CERN-PH-TH-2009/175.

TROTTER STRUCTURE

ZZ and X do not commute. Applying them sequentially is always an approximation — but the structure of that approximation depends on the order.

For two operators A and B that do not commute, the Baker-Campbell-Hausdorff identity gives:

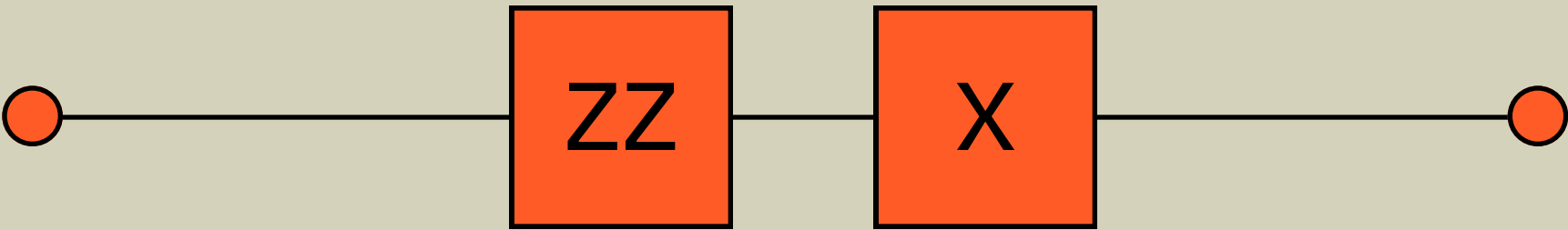
$$e^A \cdot e^B = e^{A+B+\frac{1}{2}[A,B]+\frac{1}{12}[A,[A,B]]-\frac{1}{12}[B,[A,B]]+\dots}$$

where the commutator terms contribute the error

[3] Layden, D. (2021). First-order Trotter error from a second-order perspective. arXiv:2107.08032v1 [quant-ph]. Published as Physical Review Letters, 128, 210501 (2022).

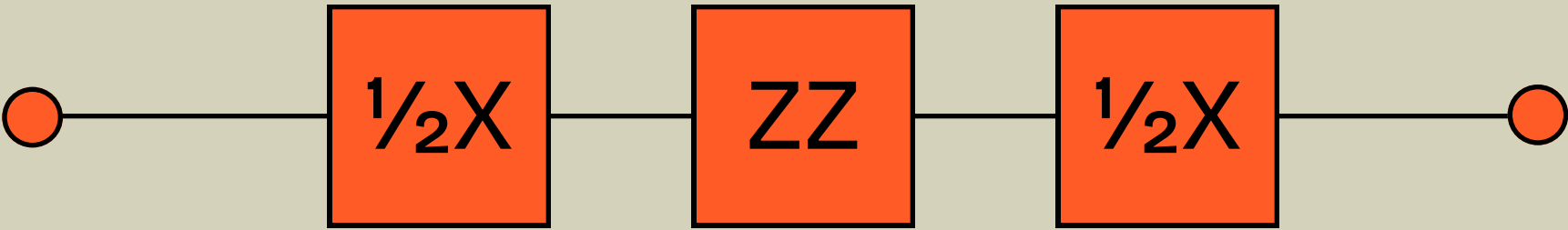
LIE-TROTTER

$$e^{-iH_{ZZ} dt} \cdot e^{-iH_X dt} = e^{-i(H_{ZZ}+H_X) dt} - \frac{1}{2}[H_{ZZ}, H_X] dt^2 + \mathcal{O}(dt^3)$$



SUZUKI-TROTTER DECOMPOSITION

$$e^{-iH_X dt/2} \cdot e^{-iH_{ZZ} dt} \cdot e^{-iH_X dt/2} = e^{-i(H_{ZZ}+H_X) dt} + \mathcal{O}(dt^3)$$



COUNTERDIABATIC DRIVING

Counterdiabatic driving adds a correction term at every step that actively suppresses the drift away from the instantaneous ground state between steps.

- Near the critical point, the system can drift into an excited state—a diabatic excitation
- The **adiabatic gauge potential** (AGP) describes in which direction the system drifts [4]

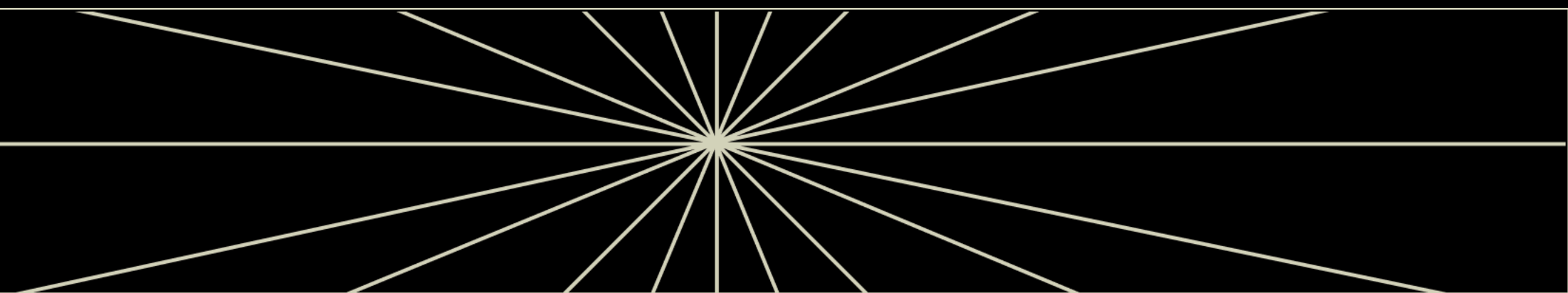
$$AGP \propto \sum_{\langle ij \rangle} (Z_i Y_j + Y_i Z_j)$$

- Adding a term proportional to the first-order AGP at every step cancels the first order drift term

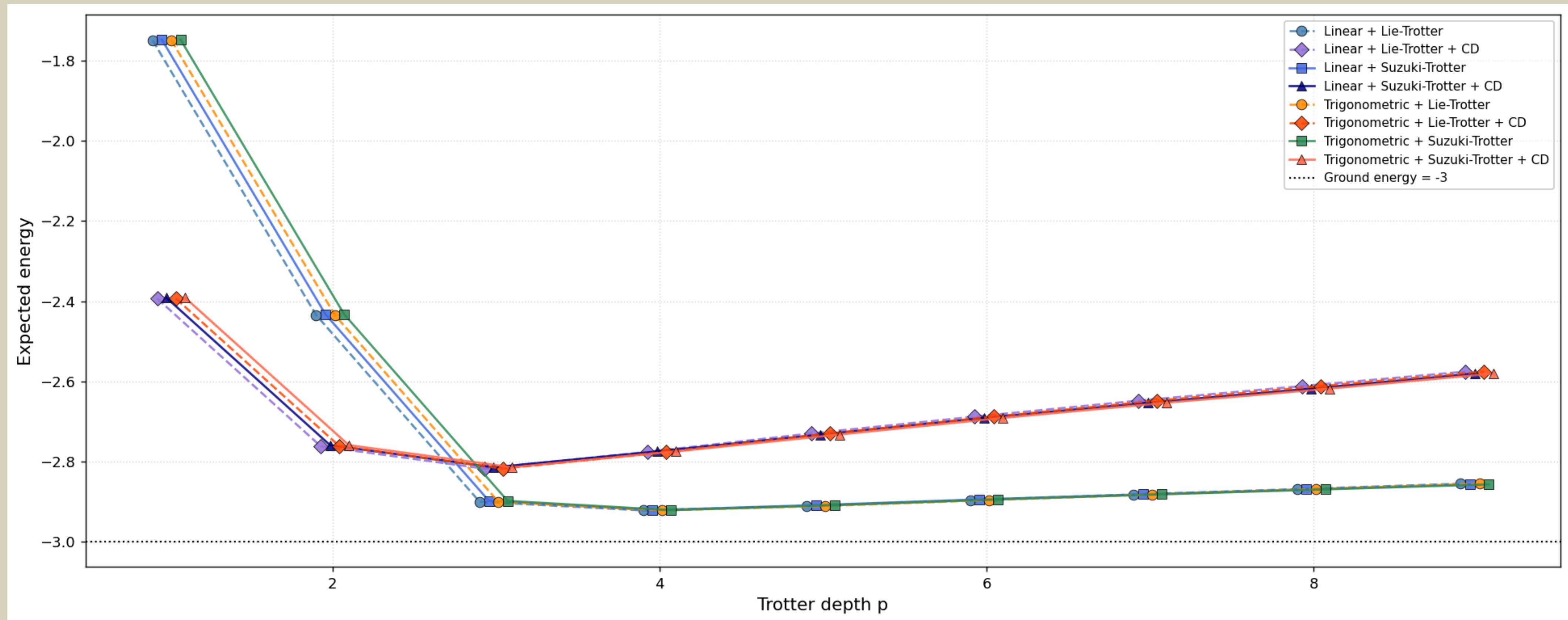
$$e^{-i\delta Y_i Z_j} = R_X\left(-\frac{\pi}{2}\right)_i \cdot RZZ(2\delta)_{ij} \cdot R_X\left(+\frac{\pi}{2}\right)_i$$

[4] Hegade, N. N., Paul, K., Ding, Y., Sanz, M., Albarrán-Arriagada, F., Solano, E., & Chen, X. (2021). Shortcuts to adiabaticity in digitized adiabatic quantum computing. Physical Review Applied, 15, 024038. arXiv:2009.03539.

RESULTS

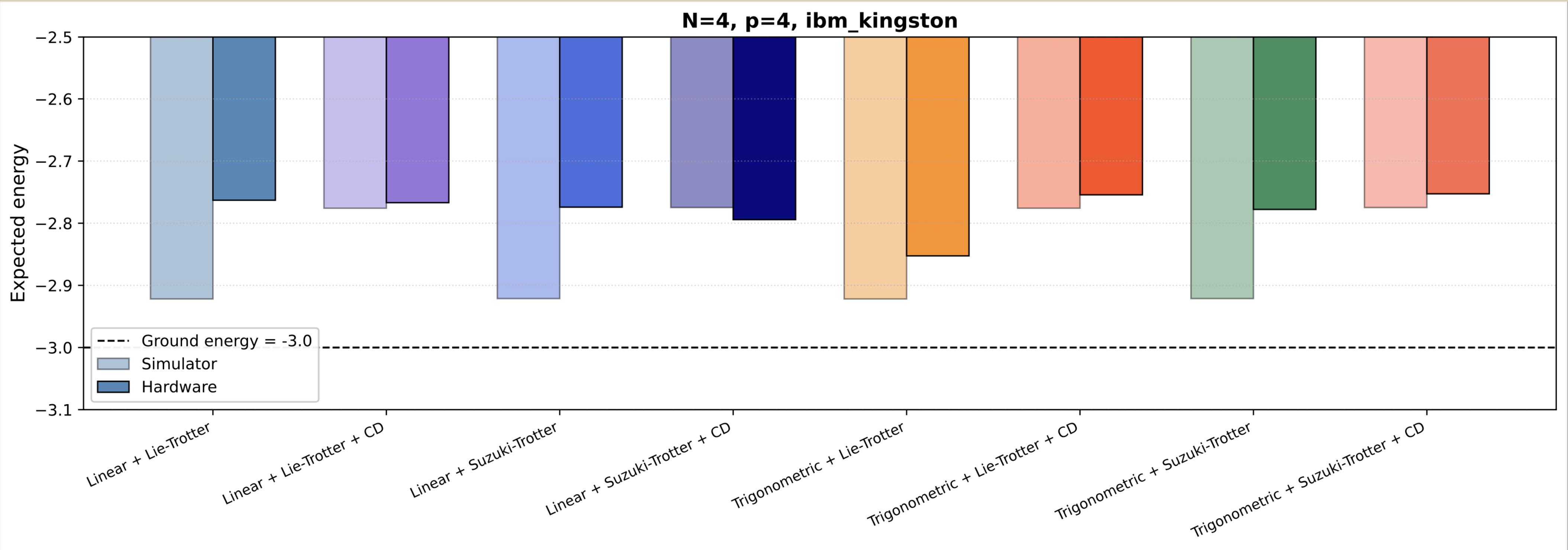


SIMULATED EXPECTED ENERGY



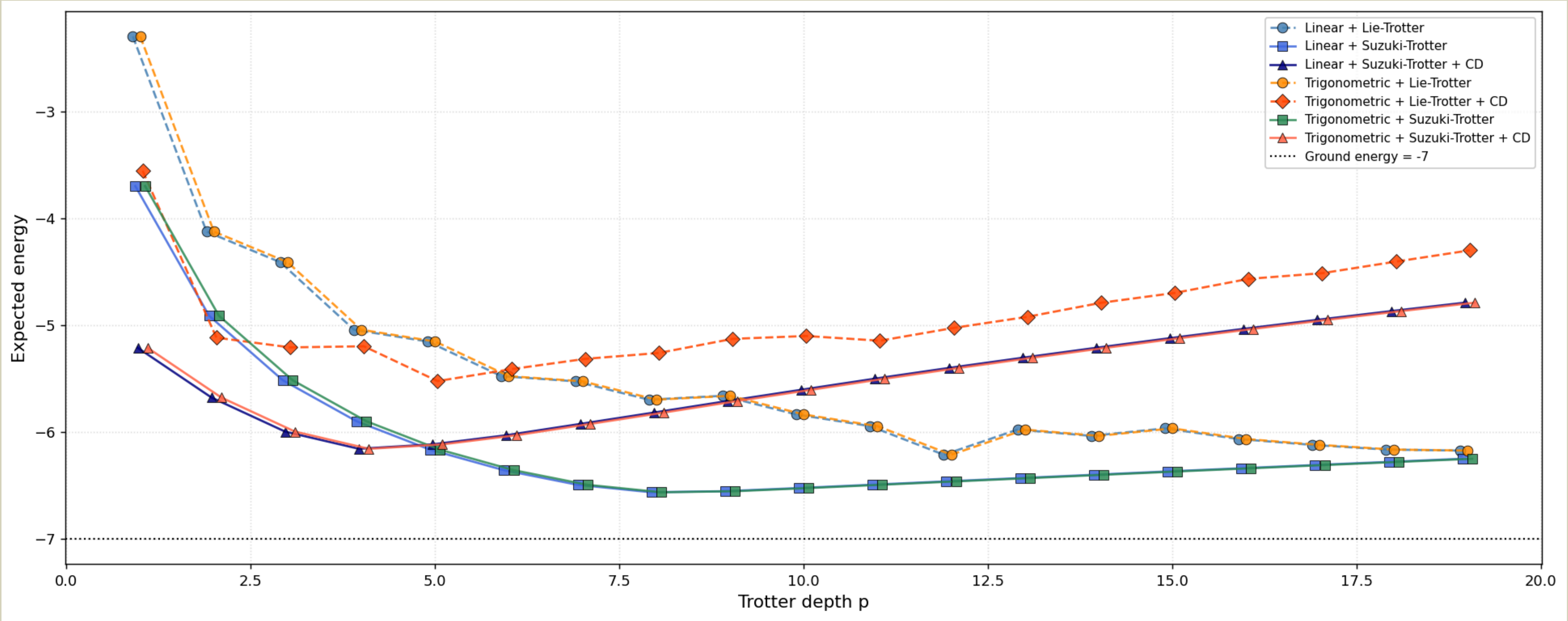
N = 4

EXPECTED ENERGY VIA HERON R2



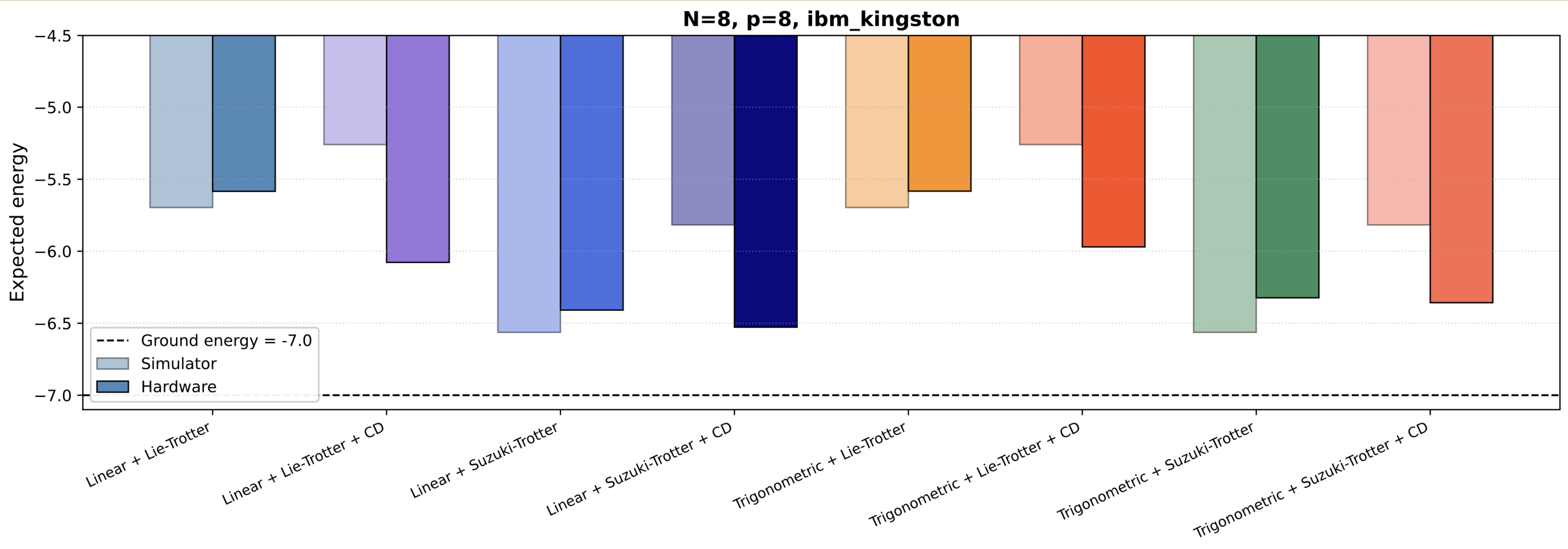
N = 4, P = 4

SIMULATED EXPECTED ENERGY



N = 8

EXPECTED ENERGY VIA HERON R2



N = 8, P = 8

KEY FINDINGS

- The best ansatz on quantum hardware is not always the best ansatz on a classical simulator
- Theoretical improvements don't always survive compilation
- Independent-error models over-penalize structured circuits
- Pulse-level compilation matters more than gate count
- Error structure matters more than error magnitude

